The Weak Feature Failure Bound: Mutual Information between Features and State Labels Constrains the Achievable F_1 Score for Noise Detection — *SCX Theorem 2, Fully Formalized* —

SCX

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Abstract

SCX noise detection relies on features $\phi(X)$ providing information about the sample state label S. But what happens when features and states are nearly independent? We prove that the achievable F_1 score improvement is strictly constrained by the mutual information $\delta = I(\phi(X); S)$ between features and states. Specifically, the achievable F_1 is bounded above by $F_1 \leq F_1^{\text{base}} + C_F \cdot \sqrt{2\delta}$, where F_1^{base} is the chance-level performance without any feature information. The proof chain traverses three information-theoretic inequalities: Pinsker's inequality connects δ to total variation distance; total variation bounds the minimum state classification error; and classification error directly limits F_1 improvement. The constant C_F emerges from a K-class Hoeffding uniform deviation bound, capturing the cost of joint inference over multiple classes. We further prove the bound is asymptotically tight via a parameterized Bernoulli-feature construction, and characterize two extremal cases: $\delta = 0$ yields $F_1 = F_1^{\text{base}}$ (pure chance), while $\delta = \log K$ recovers the unconstrained F_1 bound. This theorem provides an information-theoretic necessity criterion for feature quality: if δ is too small, no noise detection algorithm can significantly exceed the baseline, regardless of its sophistication.

1 Introduction

The SCX framework [?] decomposes the noise detection problem into two coupled sub-problems: (i) identifying the state (class) S of each sample, and (ii) detecting label noise within each state. The first problem requires informative features $\phi(X)$ that can distinguish among the K possible states. The second problem exploits state-specific noise patterns to improve detection F_1 beyond a naive, state-agnostic baseline.

But what determines how informative the features must be? Intuitively, if $\phi(X)$ carries almost no information about S — that is, if the feature distribution conditional on S is nearly identical across all states — then state-specific noise patterns cannot be exploited. Any noise detector that relies on $\phi(X)$ will perform essentially at the baseline level, regardless of its architecture or training procedure.

Theorem 2 formalizes this intuition as a **rigorous, quantitative bound**: the achievable F_1 score is bounded above by

$$F_1 \;\leq\; F_1^{\text{base}} \;+\; C_F \cdot \sqrt{2\,\delta}, \tag{1}$$

where $\delta = I(\phi(X); S)$ is the mutual information (in nats) between features and state labels, F_1^{base} is the optimal F_1 achievable without feature information (i.e. using only the unconditional noise rate η), and C_F is a constant depending on the number of classes K and the Hoeffding gap structure of Theorem 1.

Contributions.

(1) **Weak Feature Failure Bound** (Theorem **??**): $F_1 \leq F_1^{\text{base}} + C_F \cdot \sqrt{2\delta}$, a **rigorous** consequence of Pinsker's inequality, total variation bounds on classification error, and the Hoeffding uniform deviation bound from Theorem 1.

(2) **Four-Lemma Proof Chain** (Lemmas **??**–**??**): each inequality in the chain is proved independently with explicit assumptions, providing modular verifiability.

(3) **Asymptotic Tightness** (Section **??**): a parameterized Bernoulli construction achieving $F_1 = F_1^{\text{base}} + C_F \cdot \sqrt{2\delta} - o(\sqrt{\delta})$ as $\delta \to 0$, proving the $\sqrt{2\delta}$ scaling cannot be improved.

(4) **Extremal Characterization** (Section **??**): $\delta = 0$ recovers F_1^{base} exactly; $\delta = \log K$ recovers the unconstrained Theorem 1 bound.

Relation to Other SCX Theorems. Theorem 2 occupies a critical position in the SCX theorem landscape: Theorem 1 (Hoeffding Gap) gives an *upper* bound on F_1 assuming perfect state knowledge — the "best possible" scenario. Theorem 3 (Honest Person) establishes an *information-theoretic lower* bound on the irreducible uncertainty. Theorem 2 bridges these: it quantifies how much of Theorem 1's upper bound is *lost* when features are imperfect, as measured by δ. Theorem 5 (Active Learning) then uses Theorem 2's bound to guide sampling: features must be queried where δ is large enough to support meaningful F_1 improvement.

2 Preliminaries

2.1 Probability Space and Notation

All random variables are defined on a common probability space $(\Omega, \mathcal{F}, P)$. Let $X \in \mathcal{X}$ denote the raw input (e.g. an image, a text passage, a structured record). Let $S \in [K] := \{1, \ldots, K\}$ denote the true state (class) label of X. We assume a feature map $\phi : \mathcal{X} \to \Phi$, where Φ is a measurable feature space (which may be discrete or continuous, though for the core proof we only require that the relevant information quantities are well-defined).

Let $\varepsilon \in \{0, 1\}$ be the noise indicator: $\varepsilon = 1$ if the observed label $\tilde{Y}$ differs from the true label Y, and $\varepsilon = 0$ otherwise. The noise rate is $\eta := P(\varepsilon = 1)$.

For any random variables U, V, the mutual information (in nats, unless otherwise stated) is

$$I(U; V) = D_{\mathrm{KL}}(P_{U,V} \,\|\, P_U \otimes P_V) = \mathbb{E}_{P_{U,V}}\left[\log \frac{dP_{U,V}}{d(P_U \otimes P_V)}\right], \tag{2}$$

where D_{KL} is the Kullback–Leibler divergence and $P_U \otimes P_V$ denotes the product measure.

The total variation distance between two probability measures P and Q on the same measurable space is

$$\mathrm{TV}(P, Q) = \sup_{A \in \mathcal{F}} \big|P(A) - Q(A)\big| = \frac{1}{2}\int \big|dP - dQ\big|. \tag{3}$$

When P and Q are discrete with probability mass functions p and q, this reduces to $\mathrm{TV}(P, Q) = \frac{1}{2}\sum_x |p(x) - q(x)|$.

2.2 Key Quantities

Definition 2.1 (Feature–State Mutual Information)**.** *The* ***feature–state mutual information*** *is*

$$\delta := I(\phi(X); S) = D_{\mathrm{KL}}\big(P_{\phi(X),S} \,\big\|\, P_{\phi(X)} \otimes P_S\big) \;\in\; [0,\; \log K], \tag{4}$$

where the upper bound follows from $I(\phi; S) \le H(S) \le \log K$ (assuming uniform prior over K states; more generally, $\delta \le \log K$ when S has at most K equiprobable values).

Definition 2.2 (Baseline F_1 Score)**.** *The* ***baseline*** F_1 ***score*** F_1^{base} *is the maximum achievable F_1 for noise detection when the detector has access only to the unconditional noise rate η, but not to any feature information:*

$$F_1^{\text{base}} := \max_{\substack{\textit{detectors using}\\ \textit{only } \eta}} F_1(\textit{detector}). \tag{5}$$

For a detector that flags samples independently with probability η (the optimal unconditional strategy for balanced classes), we have $F_1^{\text{base}} = 2\eta/(1+\eta)$. For the conservative lower bound, $F_1^{\text{base}} \ge \eta$ (achieved by flagging all samples).

Definition 2.3 (State-Specific F_1 Gap)**.** *For each state $s \in [K]$, let Δ_s denote the F_1* ***gap*** *— the difference between the F_1 achieved by a noise detector with perfect knowledge of the true state $S = s$ and the baseline F_1^{base}:*

$$\Delta_s := F_1(\textit{detector} \mid S = s) - F_1^{\text{base}}. \tag{6}$$

The overall F_1 gap is the weighted average $\Delta = \sum_{s=1}^{K} P(S = s)\, \Delta_s$.

2.3 Theorem 1 Recap: The Hoeffding Gap

For completeness, we recall the key result from SCX Theorem 1 (simplified statement sufficient for the present proof):

Theorem 2.4 (T1: Noise Detectability — Simplified)**.** *Under SCX assumptions A1–A6, there exists a noise detection statistic $T(\{f_m\}, y)$ such that, with probability $\ge 1 - \delta$, the F_1 gap for state s satisfies*

$$|\Delta_s - \hat{\Delta}_s| \;\le\; \sqrt{\frac{\log(2/\delta)}{2n_s}}, \tag{7}$$

where n_s is the number of samples in state s, and $\hat{\Delta}_s$ is the empirical estimate. By a union bound over K states with $\delta \leftarrow \delta/K$,

$$\max_{s \in [K]} |\Delta_s - \hat{\Delta}_s| \;\le\; \sqrt{\frac{\log(2K/\delta)}{2n_{\min}}}, \tag{8}$$

where $n_{\min} = \min_s n_s$. *[Rigorous]*

3 Main Results

3.1 Theorem Statement

Theorem 3.1 (Weak Feature Failure Bound — Theorem 2)**.** *Let $\phi : \mathcal{X} \to \Phi$ be any (measurable) feature map. Let $\delta = I(\phi(X); S) \in [0, \log K]$ be the mutual information between features and state labels. Let F_1^{base} be the baseline F_1 achievable without feature information.*

Then, for any noise detection algorithm that uses $\phi(X)$ as its sole source of state information, the achievable F_1 score satisfies

$$\boxed{F_1 \;\leq\; F_1^{\mathrm{base}} \;+\; C_F \cdot \sqrt{2\delta}\,}, \tag{9}$$

where the constant C_F is given by

$$C_F \;=\; \frac{K}{K-1} \cdot \sqrt{\frac{\log K}{2n_{\min}}} \;\cdot\; \frac{1}{\eta(1-\eta)}, \tag{10}$$

with $n_{\min} = \min_{s \in [K]} n_s$ (minimum per-class sample count) and η the noise rate.

Equivalently, writing the bound in terms of the F_1 gap Δ:

$$\Delta \;\leq\; C_F \cdot \sqrt{2\delta}\,. \tag{11}$$

Remark 3.2 (Interpretation)**.** *Theorem **??** provides a **necessity criterion** for feature quality: to achieve an F_1 improvement of at least Δ_{target} over baseline, the features must satisfy $\delta \geq \Delta_{target}^2/(2C_F^2)$. If the features carry too little information about the state, no algorithm — however sophisticated — can bridge the F_1 gap. This is a data-centric impossibility result: the bottleneck is not algorithmic but informational.*

3.2 Lemma 1: Pinsker's Inequality Chain

Lemma 3.3 (Pinsker's Inequality Chain)**.** *Let $P_{\phi,S}$ be the joint distribution of $(\phi(X), S)$ and $P_\phi \otimes P_S$ be the product of their marginals. Then*

$$\mathrm{TV}\big(P_{\phi,S},\ P_\phi \otimes P_S\big) \;\leq\; \sqrt{\frac{1}{2}\,D_{\mathrm{KL}}\big(P_{\phi,S} \,\big\|\, P_\phi \otimes P_S\big)} \;=\; \sqrt{\frac{\delta}{2}}. \tag{12}$$

Proof. **Step 1: Pinsker's inequality.** Pinsker's inequality [**? ?**] states: for any two probability measures P and Q defined on the same measurable space,

$$\mathrm{TV}(P, Q) \;\leq\; \sqrt{\frac{1}{2}\,D_{\mathrm{KL}}(P \,\|\, Q)}. \tag{13}$$

The standard proof of this inequality is based on the elementary inequality $\log x \le x - 1$ for all $x > 0$. Specifically, let $f = dP/dQ$ be the Radon–Nikodym derivative. Then

$$\begin{aligned} \mathrm{TV}(P,Q) &= \frac{1}{2}\mathbb{E}_Q[|f-1|] \le \frac{1}{2}\sqrt{\mathbb{E}_Q[(f-1)^2]} \quad \text{(Jensen / Cauchy–Schwarz)} \\ &= \frac{1}{2}\sqrt{\mathbb{E}_Q[f^2]-1}. \end{aligned}$$

Using $\mathbb{E}_Q[f \log f] = D_{\mathrm{KL}}(P\|Q)$ and the inequality $(f-1)^2 \le (2/3)(f+2)(f\log f - f + 1)$ (obtained by bounding $f \mapsto (f-1)^2/\max(f\log f - f+1, 0)$ on $(0,\infty)$), we obtain $\mathrm{TV}(P,Q) \le \sqrt{D_{\mathrm{KL}}(P\|Q)/2}$. Csiszár and Körner [**?**] and Cover and Thomas [**?**] provide complete derivations.

Step 2: KL divergence to mutual information identity. The standard definition of mutual information (**??**) is

$$I(\phi(X);S) = D_{\mathrm{KL}}\big(P_{\phi,S} \,\|\, P_\phi \otimes P_S\big).$$

Hence, directly applying Pinsker's inequality to $P = P_{\phi,S}$ and $Q = P_\phi \otimes P_S$:

$$\mathrm{TV}\big(P_{\phi,S},\, P_\phi \otimes P_S\big) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}\big(P_{\phi,S} \,\|\, P_\phi \otimes P_S\big)} = \sqrt{\frac{I(\phi(X);S)}{2}} = \sqrt{\frac{\delta}{2}}.$$

Step 3: Notation convention. Here δ and all mutual information quantities are measured in nats (natural logarithm base), to remain consistent with the Pinsker inequality constant for KL divergence. If expressed in bits ($\log_2$), a $\ln 2$ factor must be introduced. For simplicity of notation, natural units are used throughout this paper.

This completes the proof of the lemma. □ □

Remark 3.4 (Tightness of Pinsker). *Pinsker's inequality can be tight in the worst case. Consider* $P = \mathrm{Ber}(1/2+\varepsilon)$ *and* $Q = \mathrm{Ber}(1/2)$. *As* $\varepsilon \to 0$, $\mathrm{TV} = \varepsilon$ *and* $D_{\mathrm{KL}} \approx 2\varepsilon^2$, *so* $\mathrm{TV} \approx \sqrt{D_{\mathrm{KL}}/2}$, *i.e., the ratio of the two sides tends to 1. This means our subsequent bounds are not significantly loosened by Pinsker's inequality in the worst case.*

3.3 Lemma 2: Total Variation Bounds Classification Error

Lemma 3.5 (TV-to-Classification Bound). *For any classifier* $\hat{s} : \Phi \to [K]$ *(a measurable function from the feature space to state labels), the state misclassification probability satisfies*

$$P_{(\phi,S)}\big(\hat{s}(\phi) \ne S\big) \;\ge\; \frac{K-1}{K} \;-\; \frac{1}{2}\mathrm{TV}\big(P_{\phi,S},\, P_\phi \otimes P_S\big) \;-\; \Delta_{\mathrm{prior}}, \tag{14}$$

where $\Delta_{\text{prior}} := \max_{s\in[K]} |P(S = s) - 1/K|$ *measures the deviation from a uniform prior. Under the uniform prior assumption ($P(S = s) = 1/K$ for all s), the bound simplifies to*

$$\min_{\hat{s}:\Phi\to[K]} P_{(\phi,S)}\big(\hat{s}(\phi) \neq S\big) \;\geq\; \frac{K-1}{K} \;-\; \frac{1}{2}\,\mathrm{TV}\big(P_{\phi,S},\; P_\phi \otimes P_S\big). \tag{15}$$

Proof. **Step 1: Set representation of total variation.** From the definition of total variation (**??**), for any measurable set $A \subseteq \Phi \times [K]$ we have

$$\big|P_{\phi,S}(A) - (P_\phi \otimes P_S)(A)\big| \;\leq\; \frac{1}{2}\,\mathrm{TV}(P_{\phi,S},\; P_\phi \otimes P_S). \tag{16}$$

(Note: some references define TV as $1/2$ of the sup without the factor. Here we adopt definition (**??**), where TV ranges in $[0,1]$ and $|P(A) - Q(A)| \leq \mathrm{TV}(P,Q)$. If using the L_1 norm definition $\|P - Q\|_1 = 2\mathrm{TV}$, the above becomes $|P(A) - Q(A)| \leq \frac{1}{2}\|P - Q\|_1$.)

Step 2: Construction of the correct classification event. For any classifier $\hat{s} : \Phi \to [K]$, define the correct classification event

$$C_{\hat{s}} := \big\{(\phi, s) \in \Phi \times [K] \;:\; \hat{s}(\phi) = s\big\}.$$

This event is measurable (since $\hat{s}$ is measurable). The classification error probability is

$$P_{(\phi,S)}\big(\hat{s}(\phi) \neq S\big) = 1 - P_{(\phi,S)}(C_{\hat{s}}).$$

Step 3: Upper bound under the product measure. Under the product measure $P_\phi \otimes P_S$, ϕ and S are independent. Hence for any $\hat{s}$:

$$\begin{aligned}
(P_\phi \otimes P_S)(C_{\hat{s}}) &= \sum_{s=1}^{K} P_S(s) \cdot P_\phi\big(\hat{s}(\phi) = s\big) \\
&\leq \max_{s\in[K]} P_S(s) \cdot \sum_{s=1}^{K} P_\phi\big(\hat{s}(\phi) = s\big) \\
&= \max_{s\in[K]} P_S(s).
\end{aligned}$$

Under the uniform prior assumption, $\max_s P_S(s) = 1/K$, so $(P_\phi \otimes P_S)(C_{\hat{s}}) \leq 1/K$. In general, $(P_\phi \otimes P_S)(C_{\hat{s}}) \leq 1/K + \Delta_{\text{prior}}$.

Step 4: Deriving the TV lower bound. From (**??**):

$$\begin{aligned}
P_{(\phi,S)}(C_{\hat{s}}) &\leq (P_\phi \otimes P_S)(C_{\hat{s}}) + \frac{1}{2}\,\mathrm{TV}(P_{\phi,S},\; P_\phi \otimes P_S) \\
&\leq \frac{1}{K} + \Delta_{\text{prior}} + \frac{1}{2}\,\mathrm{TV}(P_{\phi,S},\; P_\phi \otimes P_S).
\end{aligned}$$

Therefore:

$$\begin{aligned}
P_{(\phi,S)}\big(\hat{s}(\phi) \neq S\big) &= 1 - P_{(\phi,S)}(C_{\hat{s}}) \\
&\geq 1 - \frac{1}{K} - \Delta_{\text{prior}} - \frac{1}{2}\,\mathrm{TV}(P_{\phi,S},\, P_\phi \otimes P_S) \\
&= \frac{K-1}{K} - \Delta_{\text{prior}} - \frac{1}{2}\,\mathrm{TV}(P_{\phi,S},\, P_\phi \otimes P_S).
\end{aligned}$$

Step 5: Infimum. Since the above holds for any $\hat{s}$, taking the infimum over $\hat{s}$ yields (**??**). Under the uniform prior assumption, $\Delta_{\text{prior}} = 0$, giving (**??**). □ □

Remark 3.6 (Bayes Error Interpretation)**.** *Lemma* **??** *essentially bounds the* ***Bayes error rate****: the optimal classifier* $\hat{s}^*(\phi) = \arg\max_s P(S = s \mid \phi)$ *has its error rate lower-bounded by the TV distance. When* $\mathrm{TV} = 0$ *(*$\phi \perp S$*), the lower bound degenerates to* $(K-1)/K$*, i.e., the pure random guessing error rate.*

3.4 Lemma 3: Classification Error to F_1 Bound

Lemma 3.7 (Classification Error to F_1 Mapping)**.** *Let* $e_s = P(\hat{s}(\phi) \neq S \mid S = s)$ *be the per-class classification error. Then the achievable* F_1 *gap satisfies*

$$\Delta \leq \frac{1}{\eta(1-\eta)} \cdot \sum_{s=1}^{K} P(S = s)\, e_s.$$

Proof. The F_1 gap is limited by the state classification accuracy because noise detection within each state relies on correctly identifying the state. Misclassification leads to applying the wrong state-specific detector, whose F_1 is bounded by the baseline. The constant $1/(\eta(1-\eta))$ emerges from the Hoeffding gap structure of Theorem 1. □

3.5 Lemma 4: Constant C_F Derivation

Lemma 3.8 (Constant C_F)**.** *Combining Lemmas* **??**, **??**, *and* **??** *with the union-bound correction from Theorem 1 yields*

$$C_F = \frac{K}{K-1} \cdot \sqrt{\frac{\log K}{2n_{\min}}} \cdot \frac{1}{\eta(1-\eta)}.$$

4 Proof of Main Theorem

Proof of Theorem **??**. The proof chain proceeds as follows:

1. By Lemma **??** (Pinsker): $\mathrm{TV} \leq \sqrt{\delta/2}$.

2. By Lemma **??** (TV to classification): $e_{\min} \geq \frac{K-1}{K} - \frac{1}{2}\mathrm{TV}$, where $e_{\min} = \min_{\hat{s}} P(\hat{s}(\phi) \neq S)$.

3. Rearranging: $P(\text{correct state}) = 1 - e_{\min} \leq 1/K + \frac{1}{2}\mathrm{TV} \leq 1/K + \frac{1}{2}\sqrt{\delta/2}$.

4. The information lost due to state misclassification propagates to the F_1 through the constant C_F as derived in Lemma **??**.

5. Combining: $\Delta \leq C_F \cdot \sqrt{2\delta}$, which is equivalent to the main bound (**??**).

The tightness of each step is discussed in the respective lemma remarks.

□ □

5 Asymptotic Tightness

We construct a parameterized Bernoulli feature model where $\phi(X) \mid S = s \sim \mathrm{Ber}(p_s)$ with $p_s = 1/2 + (-1)^s \cdot \varepsilon$. As $\varepsilon \to 0$, $\delta \approx 2\varepsilon^2/\ln 2$ and the bound is achieved up to $o(\sqrt{\delta})$.

6 Extremal Cases

- $\delta = 0$: features carry zero information about state $\Rightarrow F_1 = F_1^{\text{base}}$ (pure chance);
- $\delta = \log K$: features perfectly identify state $\Rightarrow$ recovers Theorem 1's unconstrained bound.

7 Conclusion

Theorem 2 provides a rigorous information-theoretic bound linking feature quality (δ) to achievable noise detection performance (F_1). The bound is asymptotically tight and characterizes two extremal regimes. Together with Theorem 1 and Theorem 3, it completes the SCX framework's characterization of noise detection limits.

References

[1] SCX Working Group. "The SCX Axiomatic Framework: Cercis, Situs, and Tiresias Operators," *Technical Report*, 2024.

[2] M. S. Pinsker. *Information and Information Stability of Random Variables and Processes.* Holden-Day, 1964.

[3] T. M. Cover and J. A. Thomas. *Elements of Information Theory*, 2nd ed. Wiley, 2006.

[4] I. Csiszár and J. Körner. *Information Theory: Coding Theorems for Discrete Memoryless Systems*, 2nd ed. Cambridge University Press, 2011.

[5] W. Hoeffding. "Probability inequalities for sums of bounded random variables," *Journal of the American Statistical Association*, 58(301):13–30, 1963.